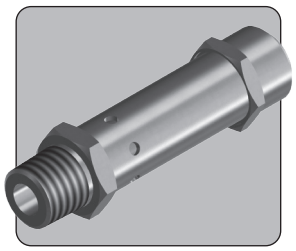
BS50

SAFETY RELIEF VALVE

1/2″

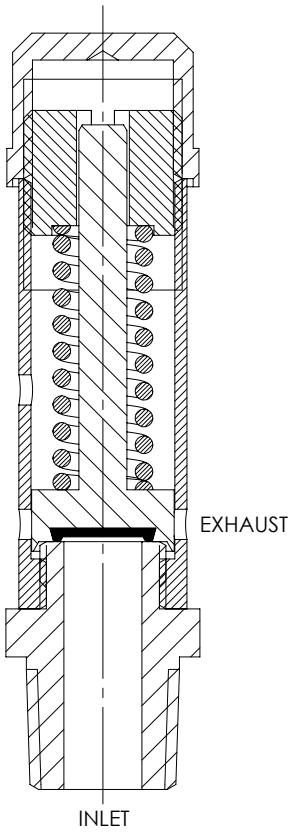
UP TO 40 BAR 580 PSI



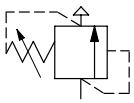
The BS50 is a vent to atmosphere safety relief valve. Designed with an elastomeric seat seal the valve combines excellent leakage performance with a high discharge flow capacity from its pop-open characteristic.

The valve is approved as a safety accessory to category IV of the Pressure Equipment Directive, and is manufactured primarily from brass to offer cost effective pressure protection.

- Approved to category IV of the Pressure Equipment Directive (2014/68/EU)
- Pop-open characteristic for maximum discharge capacity
- Vents to atmosphere – no need for exhaust side pipework
- Available preset and internally adjustable. Wire locking and tagging available on request
- Available with relieving gear
- Virtually zero leakage at 90% of set pressure
- Reseat within the greater of 5 psi or 7.5% of set pressure when valve has opened and not popped
- Reseat within 50% of set pressure once popped
- Repeatability within the greater of 2 psi or ±3% of nominal set pressure



Specifications

BASIC MODEL NUMBER	BS50
SYMBOL	 INLET
MAX WORKING PRESSURE	40 bar (580 psi)
ORIFICE SIZE	Ø9.8mm (Ø0.386″)
FLUID	Liquids and Gases See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/2″
WEIGHT	0.25 kg (0.55 lb)

Specifications may change without notice

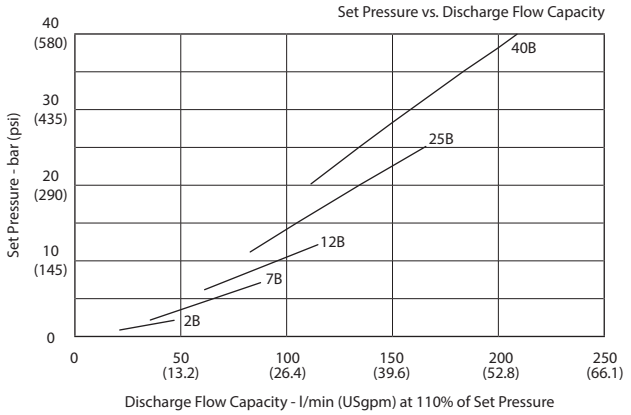
Materials

Externally Exposed and Internally Wetted Parts: Brass, 316 and 302 stainless steel.

The standard valve is designed for use with air and plain water.
Note that use with water above 100°C is not recommended.
Other media compatible with the materials of construction may be used, although consideration must be given to the safety of such fluids when vented to atmosphere.

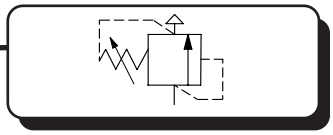
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

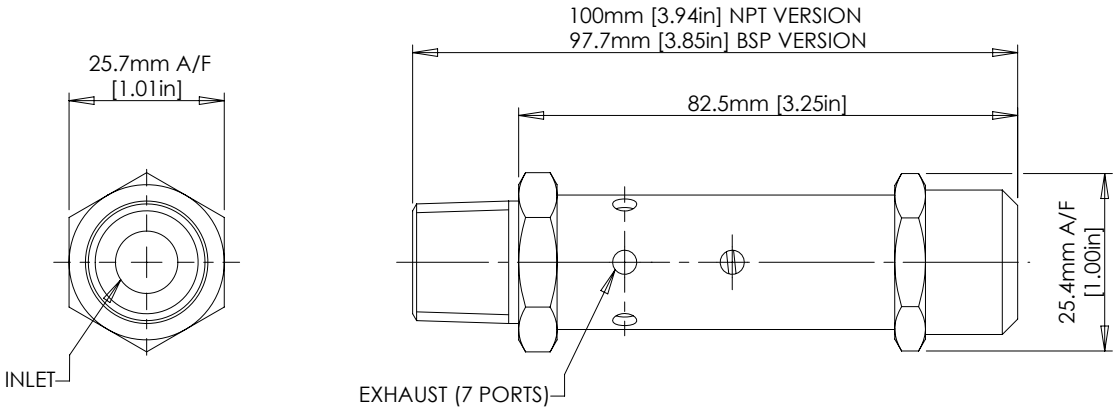
Typical Performance

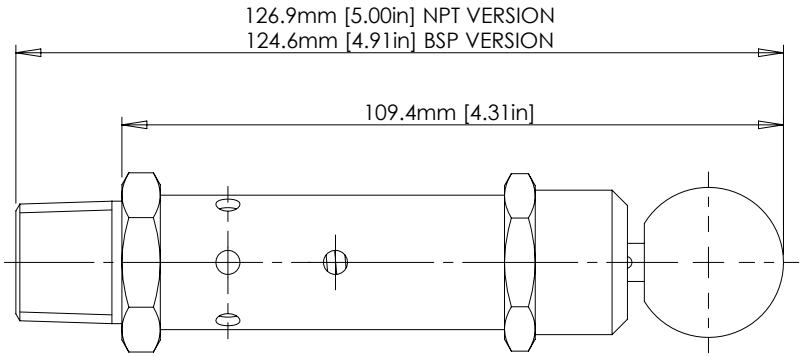


Typical performance based on water
Contact us with system details for an application specific calculation

Installation Diagram







RG - RELIEVING GEAR OPTION

Product Selector

BS50 P - RG - EP - 2B

VALVE TYPE	PORTING OPTIONS	VALVE OPTIONS	SEAL OPTIONS	SET PRESSURE RANGE
BS50	P 1/2" BSPP male N 1/2" NPT male	Leave blank if not required RG Relieving gear ⚠ Exposure to high and low fluid temperatures will influence handle temperature. Operator safety must be considered	Leave blank for Viton® seals -10˚C to +160˚C EP EPDM seals -45˚C to +150˚C NBR Nitrile seals -30˚C to +105˚C Further seal options are available on request ⚠ The safety of venting high and low temperature fluids to atmosphere must be considered	2B 0.7 - 2 bar (10 - 29 psi) 7B 2 - 7 bar (29 - 102 psi) 12B 6 - 12 bar (87 - 174 psi) 25B 11 - 25 bar (160 - 363 psi) 40B 20 - 40 bar (290 - 580 psi) Advise us of set pressure if factory pre-setting is required

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK